For teachers only. Do not give this part to students.

Distribution and evidence preservation

1. Following proposal 4.1 and 4.3.2, these worksheets use the same scenarios, concepts, prompts and static data, before formal teaching. Keep weekly worksheets separate from the formal 10-item SPSAI and 4-item CKT pre- and post-tests.
2. Give out A first. Save the original observation, hypothesis, reason, variables, assumptions and setup design. Keep a copy students can consult during reflection.
3. Give out B after saving the original design. Static images replace interaction for the same four setups and four times. Students complete the readings, calculations, graph and conclusions.
4. Keep first readings legible and record revisions separately. Give out C only after the original investigation answers are fixed; do not reveal the learning points earlier.
5. Students consult their original hypothesis and reason in A and complete the same reflection in C. D and the full pack are for teachers only; distribute the parts separately.
6. Use the same situation, questions, hints, data, language and available time for both groups. Static reading and digital interaction are different treatments; paper completion is not a measure of operating skill.
7. The current VL2 has no extension investigation; the paper does not add one. Keep formal research pre- and post-tests separate from these activity worksheets.
8. Test-print B and C in grayscale before use. Check liquid surfaces, water traces, rulers and scientific labels. Time, clicks and completion rate do not directly become SPS scores.

The activity scoring guide is a draft for researcher calibration, not a validated research instrument.

Automatic scoring in the shared guide refers to VL Excel. Teachers mark paper answers using the same criteria; handwriting is not read automatically.

Blank teacher scores mean awaiting marking, not zero. The six SPS areas total 24 points; new knowledge totals 8.

For teachers only. Do not give this part to students.

Reference answers

Independent variable	Light intensity (光強度)
Dependent variable	Height / average upward speed of the red water trace
Controlled variables	Temperature / Humidity (濕度) and air movement / Red water concentration and depth of immersion (浸入深度) / Total leaf area of the setups with leaves / Length and thickness of the celery stalks
What assumptions does this investigation make?	Changing the lamp distance does not make the setups have different temperatures. Setups A, B and C have the same total leaf area. Humidity (濕度) and air movement are the same for all setups.
1. How is light intensity (光強度) related to the upward speed of the water trace?	increase
2. What is the role of leaves?	faster
2. What is the role of leaves?	help water move upwards
3. What can this experiment tell us?	The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率).

min	A (cm)	B (cm)	C (cm)	D (cm)
0	0.0	0.0	0.0	0.0
10	4.0	2.5	1.5	0.5
20	8.0	5.0	3.0	1.0
30	12.0	7.5	4.5	1.5
cm/min	0.40	0.25	0.15	0.05

For teachers only. Do not give this part to students.

References for written answers and assumptions

After some time, the red water trace extends upwards in some parts inside the celery stalk.

There is no single answer. Your hypothesis should be testable (可測試的) by changing light intensity (光強度) and measuring the upward speed of the water trace.

Explain the link you expect between light and water transport. Your prediction does not have to match the final result.

Compare setup D without leaves with setup B with leaves. Both have a lamp distance of 20 cm and the same other conditions. The main difference is whether leaves are present.

Materials: 4 celery stalks, 4 desk lamps, 4 cups of red water, 4 rulers of 30 cm, 1 pair of scissors and 1 timer. Set the lamp distances for the three celery stalks with leaves to 10, 20 and 30 cm. Use the same lamp distance for the setup without leaves and one setup with leaves. Immerse each stalk 2 cm into the red water.

Changing the lamp distance does not make the setups have different temperatures.

Temperature can also affect transpiration (蒸騰). This simulation keeps all setups at 25°C so light and temperature do not change together.

Setups A, B and C have the same total leaf area.

Leaf area can affect water loss, so setups A, B and C with leaves should have the same total leaf area. Setup D without leaves provides a separate comparison.

Leaf area does not affect the results, so it does not need to be kept the same.

Leaf area can affect water loss. Keep it the same when comparing light intensity (光強度).

Humidity (濕度) and air movement are the same for all setups.

Humidity (濕度) and air movement can also affect transpiration (蒸騰). Keep them the same when comparing light intensity (光強度).

For teachers only. Do not give this part to students.

Scoring guide

The activity scoring guide is a draft for researcher calibration, not a validated research instrument.

Category / item	Maximum	Scoring method	Full credit	Partial credit	Zero credit
First observations	2	Teacher marks	Describe the red trace extending upwards in parts of the leaf stalk; separate observation from explanation.	1: Mention the trace or colour change but omit its position or direction.	No answer, unrelated answer, or report an unobserved mechanism as a direct observation.
Ruler readings	2	Automatic	All 16 final confirmed readings are within 0.25 cm of the ruler model.	2 × number of correct readings ÷ 16, to two decimal places.	No correct confirmed readings.
Variable classification	4	Automatic	Independent: light intensity (光強度) (1); dependent: trace height / average upward speed (1); five controlled variables (2).	Controlled variables: 0.4 each. Selecting the independent or dependent variable as controlled gives zero for that column.	Not answered or incorrectly classified.
Testable (可測試的) hypothesis and reason	2	Teacher marks	Predict how changing light intensity (光強度) affects trace speed and give a relevant reason. The prediction need not match the result.	1: A testable (可測試的) hypothesis with an incomplete or unclear reason.	No testable (可測試的) hypothesis, or copy a conclusion without a prediction made before the experiment.

Automatic scoring in the shared guide refers to VL Excel. Teachers mark paper answers using the same criteria; handwriting is not read automatically.

For teachers only. Do not give this part to students.

Scoring guide

The activity scoring guide is a draft for researcher calibration, not a validated research instrument.

Category / item	Maximum	Scoring method	Full credit	Partial credit	Zero credit
Experiment assumptions	1	Automatic	Select only unchanged temperature, equal total leaf area in the leafy setups, and equal humidity (濕度) and air flow.	Credit requires the whole set to be correct; no partial credit.	An extra, missing or absent selection.
Control setup design	1	Teacher marks	Compare with and without leaves at the same light distance, keeping other conditions the same.	This item receives 0 or 1 only.	Other factors are not kept the same, or no answer.
Four complete measurement rounds	2	Automatic	Valid confirmed readings for all four setups at 0, 10, 20 and 30 min.	0.5 per complete round. Reading accuracy is scored under observing, not counted twice.	No complete confirmed measurement rounds.
Experiment setup plan	2	Teacher marks	Drawing or words show four stalks: three leafy light distances and a leafless control at the same distance; include red water and ruler / timing arrangements.	1: A workable plan missing one main arrangement.	Unworkable, unrelated or incomplete.

Automatic scoring in the shared guide refers to VL Excel. Teachers mark paper answers using the same criteria; handwriting is not read automatically.

For teachers only. Do not give this part to students.

Scoring guide

The activity scoring guide is a draft for researcher calibration, not a validated research instrument.

Category / item	Maximum	Scoring method	Full credit	Partial credit	Zero credit
Average upward speed calculations	2	Automatic	For each setup: (30 min reading − 0 min reading) ÷ 30. Each correct setup earns 0.5.	Use the student’s final readings. New records use two decimal places and a 0.0051 cm/min rounding tolerance; old records retain three decimal places and 0.00051 tolerance.	No correct calculations. Do not substitute model readings when assessing arithmetic.
Four conclusion choices	2	Automatic	Greater light intensity (光強度) usually gives faster movement; B/D comparison matches student calculations; leaves help transport; the trace is indirect evidence.	0.5 per correct choice. The B/D comparison needs sufficient data.	Incorrect or unanswered.
Graph points and curve	2	Automatic	Four points: X = each setup’s lux, Y = the student’s calculated speed (0.375 each); correctly join A/B/C (0.5).	Calculate with unrounded point credit, then round to two decimals. Do not include D in the leafy curve.	No correct points or curve. Calculation errors are scored separately; assess the graph using the student’s calculations.
Written communication	2	Teacher marks	Observations, setup plan and reflection communicate ideas clearly, using clear data, setup names and units. Assess expression separately from conceptual accuracy.	1: Understandable but vague or missing necessary labels.	Cannot be understood or no assessable communication.

Automatic scoring in the shared guide refers to VL Excel. Teachers mark paper answers using the same criteria; handwriting is not read automatically.

For teachers only. Do not give this part to students.

Scoring guide

The activity scoring guide is a draft for researcher calibration, not a validated research instrument.

Category / item	Maximum	Scoring method	Full credit	Partial credit	Zero credit
Reflection: transpiration (蒸騰) and stomata (氣孔)	2	Teacher marks	The revised explanation correctly describes evaporation (蒸發) from mesophyll water films (水膜) and water vapour (水汽) leaving through stomata (氣孔) by diffusion.	1: Correctly mention leaf evaporation (蒸發) or vapour loss through stomata (氣孔), but the mechanism is incomplete.	No use of the concept or a core misconception.
Reflection: xylem (木質部) and transpiration pull (蒸騰拉力)	2	Teacher marks	Link transpiration (蒸騰) to transpiration pull (蒸騰拉力) and upward water transport through xylem vessels (木質導管).	1: Correctly mention xylem (木質部) transport or transpiration pull (蒸騰拉力) without linking them.	No use of the concept or a core misconception.
Reflection: effect of light intensity (光強度)	2	Teacher marks	Link stronger light to wider stomatal opening (氣孔張開), transpiration (蒸騰) and upward transport. Do not equate trace speed with transpiration rate (蒸騰速率).	1: Correctly describe stronger light and faster movement but omit the mechanism.	Reverse the relationship, directly equate trace speed with transpiration rate (蒸騰速率), or do not use the concept.
Reflection: revising the original explanation	2	Teacher marks	Judge whether the original hypothesis is supported by the student’ s data, then revise or improve the explanation using data and learning points. A correct initial hypothesis can earn full credit.	1: A judgement or revision without a clear data or concept link.	Only copy learning points, do not test the original hypothesis, or do not submit reflection.

Automatic scoring in the shared guide refers to VL Excel. Teachers mark paper answers using the same criteria; handwriting is not read automatically.

For teachers only. Do not give this part to students.

Scoring guide

The activity scoring guide is a draft for researcher calibration, not a validated research instrument.

Category / item	Maximum	Scoring method	Full credit	Partial credit	Zero credit
Totals and completion requirements	32	Formula totals	Six SPS areas × 4 = 24; four knowledge areas × 2 = 8; total 32. Show the overall score only after all nine teacher scoring cells and the reflection are complete.	Blank means awaiting marking, not zero. Excel recalculates on opening. Save the workbook after marking.	A new download does not import scores from an earlier workbook. Teacher scores are kept only in the saved Excel file.
Colours and research use	0	Information	Background colours distinguish SPS and knowledge. Green = correct / full credit; red = incorrect / zero; orange = partial; awaiting marking uses a neutral colour.	Written-answer cells change colour when marked. PDFs show no scores; activity records are not SPS scores.	This is a VL2 draft. Align criteria, weights and raters across the four modules before longitudinal comparisons; it is not a validated SPSAI replacement.

Automatic scoring in the shared guide refers to VL Excel. Teachers mark paper answers using the same criteria; handwriting is not read automatically.